Controller, Sensor and RTC Bring-Up

S.C.O.U.T. Electrical Subsystem · Test Report

Report number	SCOUT-E-TR-001, Rev C	**Status**	Draft for review
Test date	2026-10-29, 12:15–13:30	**Location**	Room 4020, ECEN Soldering Lab
Tested by	Isabella Rodriguez	**Reviewed by**	Dr. Shoba Krishnan; Dr. Jes Kuczenski
Test plan	Bench Test Plan 2026-10-24, Tests 1, 3, 3b, 4, 11	**Tracking**	SCO-152, SCO-60, SCO-204
Overall result	**PASS** · 5 of 5 tests passed · 1 incident found and closed (AR-01)		

1. Purpose and scope

First powered verification of the buoy controller stack and its first two sensors: confirm the soldered hardware works, each device communicates on its intended interface, and the sensors respond correctly to physical stimuli before radio, storage and power testing.

Not tested this session: LoRa radio (no antenna fitted yet), microSD and coin-cell clock backup (parts pending), rail voltage and current draw (no multimeter), turbidity sensor (parts in transit), absolute accuracy against a calibrated reference.

2. Test setup

Hardware under test	Feather M0 RFM95 (3178) with Adalogger FeatherWing (2922) stacked on hand-soldered headers; DS18B20 waterproof probe (381) on D12 with 4.7 kΩ pull-up; SHT40 (4885) on I²C via STEMMA QT cable (4209). Powered from laptop USB, 3.3 V from the Feather regulator.
Wiring	Per Electrical Block Diagram (2026-10-28) on a breadboard. Probe: red → 3V3, blue → GND, yellow → D12. SHT40: red 3V3, black GND, blue SDA, yellow SCL. No differences from the diagram.
Software	Arduino IDE 2.3.10; Adafruit SAMD 1.7.17; OneWire 2.3.8, DallasTemperature 4.0.6, Adafruit SHT4x 1.0.5, RTClib 2.1.4. Bench test sketches `scout_t1_hello`, `t3_ds18b20`, `t3_diag`, `t11_sht40`, `t3b_temp_leak`, `t4_rtc` (not the flight firmware; same pin assignments).
Test equipment	Laptop (power, upload, logging). No multimeter or calibrated reference available.
Environment	Indoor lab, still air, ≈ 22.5 °C and ≈ 33.5 % RH (from the sensors under test).

3. Results summary

ID	Test	Requirement	Pass criterion	Measured	Result
TC-01	Controller bring-up	Enabling	Uploads; serial output at 1 Hz; banner after reset	11,020 B sketch uploaded; 1 Hz output; banner after reset	**PASS**
TC-02	Water-temperature probe (DS18B20)	N1, M1, M2	1 device found; room ±1 °C; tracks warming and cooling; no dropouts	22 °C room; rise to 27.25 °C, smooth decay; 0.0625 °C steps; 0 dropouts after fix	**PASS**
TC-03	Leak sensor (SHT40)	N10, N12, M19	Within 0.7 °C of probe; RH steady ±1 %; clear step; recovers to ±1 %	ΔT 0.39 °C; 33.6 % RH steady; 33 → 96 % on breath; recovered in 90 s	**PASS**
TC-04	Both sensors together	N1, N12	Both report every cycle; each stimulus moves only its own sensor	35 cycles, no errors; probe +4.3 °C with SHT40 unchanged; SHT40 AH ×3.9 with probe unaffected	**PASS**
TC-05	Real-time clock (PCF8523)	N8, M13	Found on I²C; counts 1 s; keeps time through reset	Set to 13:23:18, counted at 1 Hz; continued after reset	**PASS***

PASS: all criteria met. PARTIAL: shown under easier conditions than the requirement. FAIL: a criterion not met. * On USB power; coin-cell backup and drift not yet tested. Requirement IDs are the needs (N) and metrics (M) in the Project Specifications.

4. Test details

TC-01 Controller bring-up · PASS

Procedure. Selected "Adafruit Feather M0 (SAMD21)", uploaded `scout_t1_hello`, opened the Serial Monitor, pressed reset once.

Observations. Compiled to 11,020 B (4 % of flash); counter printed at 1 Hz; start-up banner printed after reset; on-board LED blinked. The USB port moved from COM3 to COM5 after the first upload, which is normal for this board.

TC-02 Water-temperature probe (DS18B20) · PASS

Procedure. Wired with power off; touch-checked the tip for heating after power-up; ran `scout_t3_ds18b20` (12-bit, one reading every ≈ 2.75 s); warmed the tip by hand, released, and logged the cool-down.

Observations. First run reported no probe (AR-01, §5). After the fix: one device found, 22 °C at room temperature, tip not warm. Warming raised the reading to at least 27.25 °C; it then fell smoothly to 25.06 °C over ≈ 91 s with no dropouts. Every reading lands on a 0.0625 °C step, confirming 12-bit resolution (M2). The still-air time constant is ≈ 150–170 s; it will be much shorter in water.

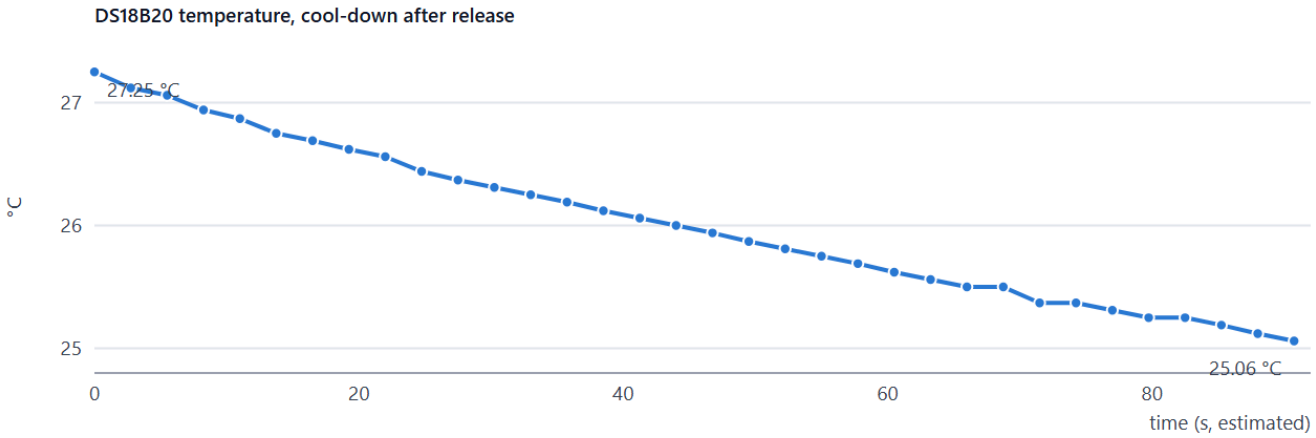


Figure 1. Probe cool-down after hand warming (time axis at 2.75 s per reading).

TC-03 Leak sensor (SHT40) · PASS

Procedure. Connected by STEMMA QT cable with power off; ran `scout_t11_sht40` (heater off, one reading every 10 s); logged 1 minute at rest; reset, logged 3 baseline readings, breathed on the sensor for ≈ 10 s, then left it alone.

Observations. Found on the first attempt with the probe still connected. At rest: 22.82–22.86 °C and 33.55–33.58 % RH, 0.39 °C above the probe, within the two sensors' combined tolerance (±0.7 °C). Breath raised humidity from 33.4 % to 96.1 % within 30 s, and it recovered to within 1 % of baseline 90 s after the peak.

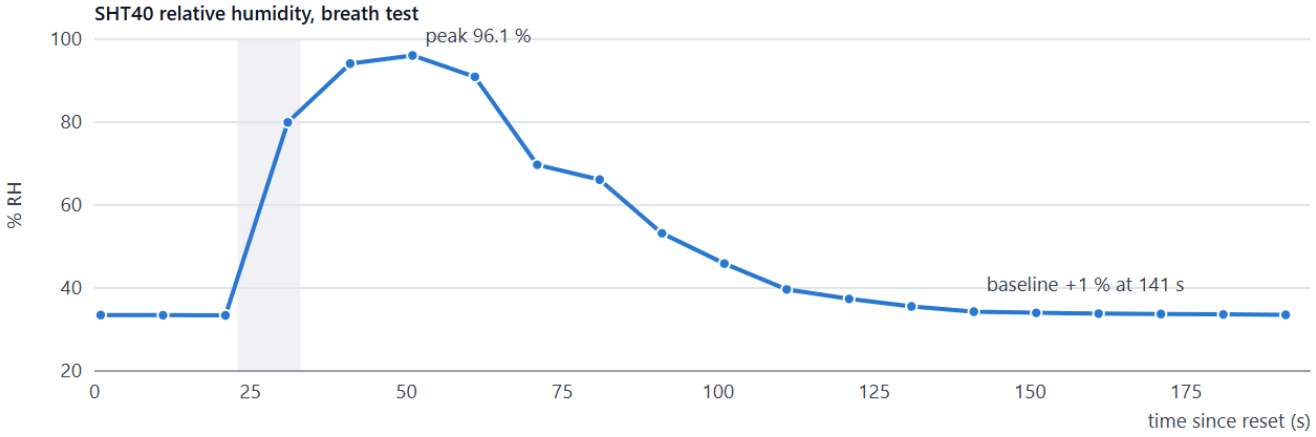


Figure 2. Relative humidity during the breath test (shaded: breath applied).

TC-04 Both sensors together · PASS

Procedure. Ran `scout_t3b_temp_leak` (both sensors plus absolute humidity, one line every ≈ 5.6 s); warmed the probe by hand, then breathed on the SHT40, then left both alone.

Observations. Warming the probe raised it from 22.37 to 26.62 °C while the SHT40 stayed at 22.7 °C and 33.7 %. Breathing on the SHT40 raised absolute humidity from 6.8 to 26.8 g/m³ while the probe kept cooling undisturbed. Absolute humidity recovered faster than relative humidity because the air was also cooling, which supports using absolute humidity for the leak alarm (SCO-204).

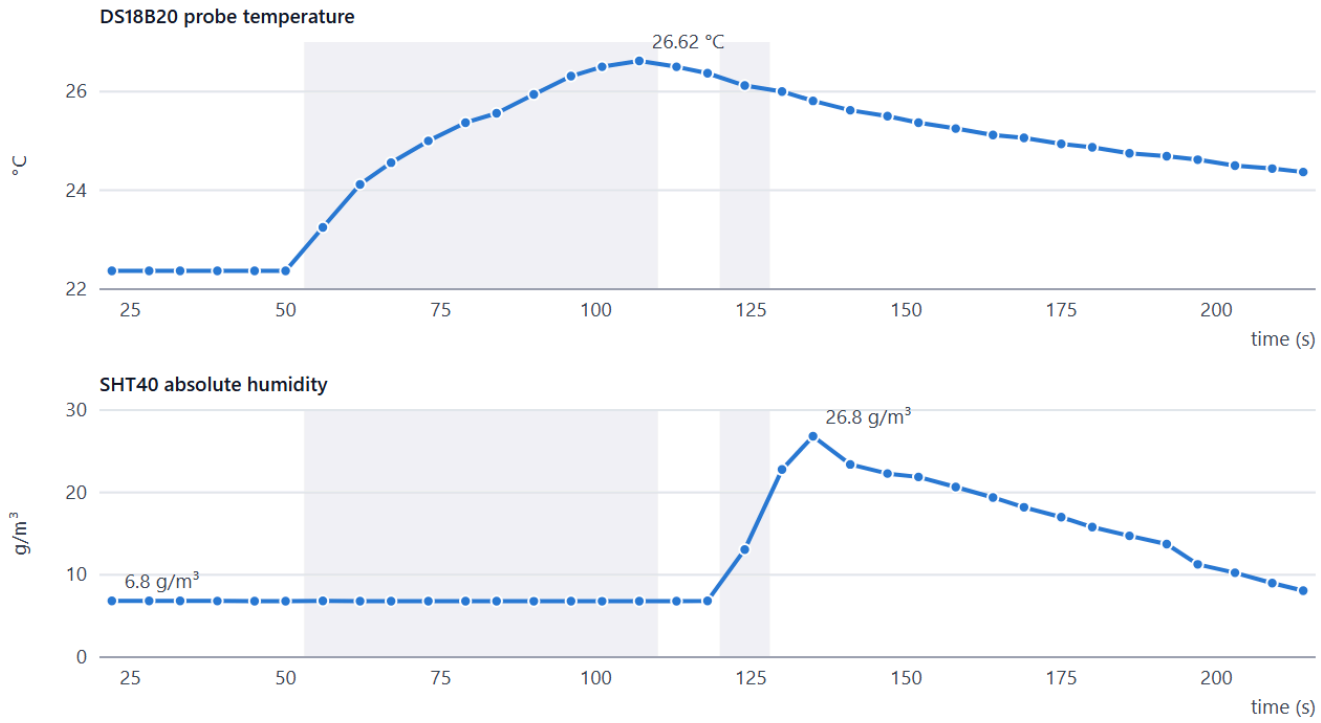


Figure 3. Concurrent test on one time axis: probe temperature (top), SHT40 absolute humidity (bottom). Shaded: probe warmed (left), breath on SHT40 (right).

TC-05 Real-time clock (PCF8523) · PASS (USB power)

Procedure. Ran `scout_t4_rtc`, watched the time count, pressed reset once.

Observations. Clock found on I²C alongside the SHT40, which also confirms the stacking-header joints. With no coin cell it started unset, as expected, and was set to 13:23:18; it counted at 1 Hz and kept time through the reset.

5. Issues and deviations

ID	Type	Description and root cause	Action taken	Status
AR-01	Incident	Probe not detected on first run ("NO PROBE −127"). A diagnostic sketch plus a wiggle check traced it to a loose jumper-to-jumper junction on the yellow data line.	Junction re-seated; no dropouts in the next 69 readings. Use a continuous or soldered wire in the final build.	Closed
D-01	Limitation	No multimeter: 3.3 V rail and current draw not measured.	Measure when a meter is available (M15).	Open
D-02	Limitation	No calibrated reference: accuracy judged by agreement between the two sensors only.	Ice-point and reference check (M1).	Open
D-03	Deviation	Bench sketches used instead of the flight firmware; clock set to local time, not UTC.	Firmware integration test; UTC set with the clock tool.	Open

6. Conclusions and next steps

- The controller, water-temperature probe, leak sensor and clock all work as assembled, individually and together, on USB, 1-Wire (D12) and I²C (0x44, 0x68).

- Verified: water-temperature resolution (M2). Partly verified: temperature accuracy (M1), clock (M13), leak sensing (M19).
- Next: microSD test (128 MB card); coin-cell backup and 24 h drift (CR1220, in transit); antennas and the shore-station radio wires, then the first buoy-to-shore packet; ice-point check; current draw once a multimeter is available.

7. Approval

Role	Name	Signature	Date
Test engineer	Isabella Rodriguez		
Reviewer	Dr. Shoba Krishnan		
Reviewer	Dr. Jes Kuczenski		

Appendix: evidence

This folder (`hardware/test/evidence/rr-25-2026-10-29/`, repo record RR-25): `screenshots/` (upload, "NO PROBE" before the fix, wiring photo, SHT40 baseline and breath test, clock) and `data/` (cool-down, SHT40 baseline, breath test and concurrent-test readings as CSV). The bench sketches are in `hardware/test/bench-sketches/`. The full-detail Rev B of this report is kept outside the repo.

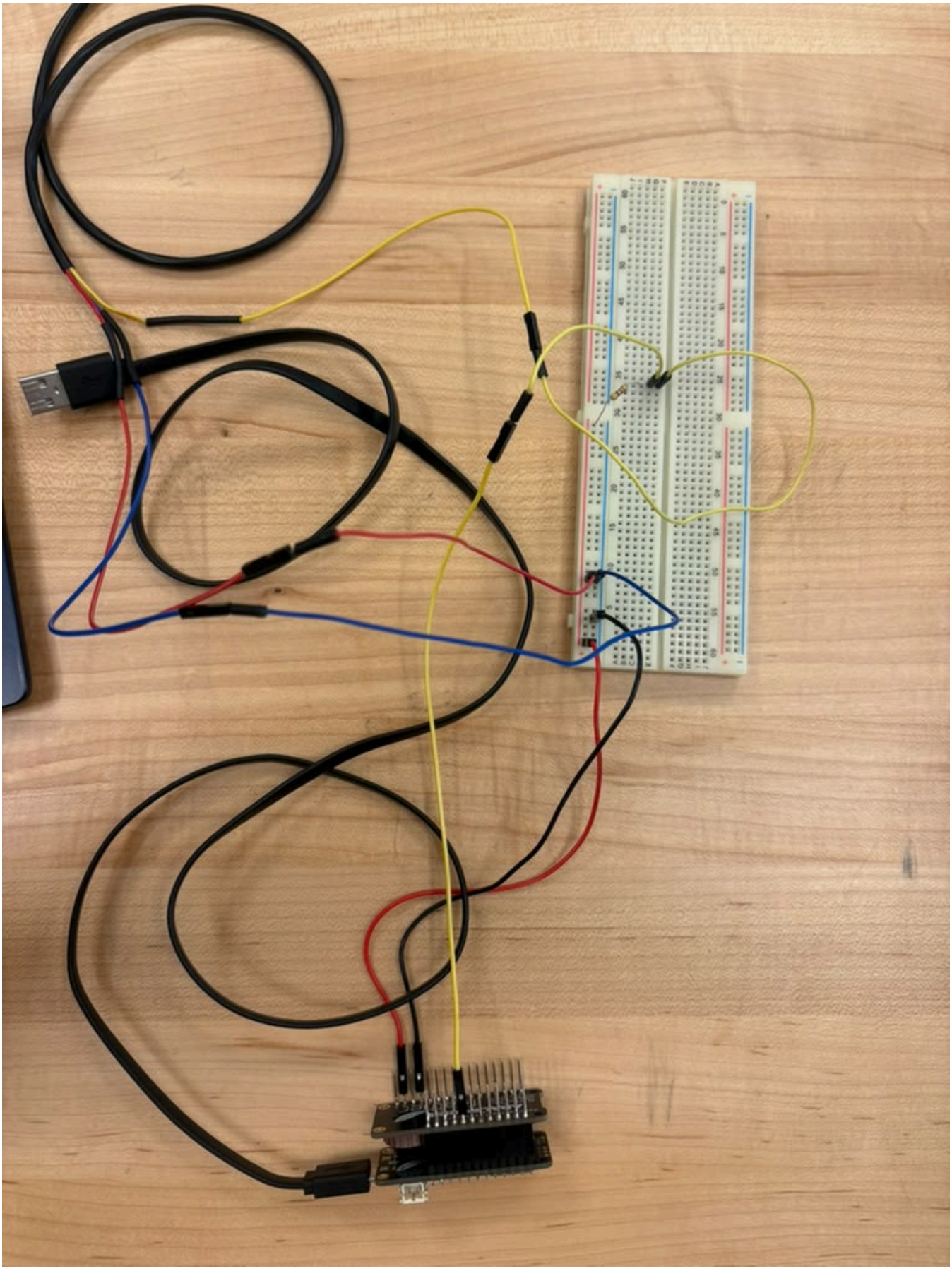


Figure 4. Bench setup: stacked M0 and Adalogger, breadboard rails, probe splices and SHT40.